

components, simplifying design and lowering the bill of materials (BOM) cost. Engineers can control LED intensity and manage LED driving without using the CPU, reducing software complexity and improving efficiency.

Applications. Suitable for applications including lighting, power supplies, battery charging, motor control, and other general purposes

Link. <https://www.electronics-foru.com/electronics-projects/wearable-heart-rate-monitor-reference-design>

OEM Brand. Microchip

Heart-rate monitor wrist band

Compact devices are now expected to track heart rate, monitor blood oxygen, and run all day without frequent charging. Designing such solutions means fitting multiple functions into a low-power package while maintaining accuracy and



battery performance, particularly since most devices spend over 90% of their time in standby.

This wristband reference design integrates a microcontroller, low-power heart rate and pulse oximeter front end, power management, and signal processing. With only a few LEDs and sensors, it delivers a complete optical heart rate solution while conserving energy. The result is longer battery life, accurate readings, and a comfortable form factor, while power management supports multiple voltage requirements and safe battery charging.

Applications. Enables efficient, reliable, and accurate wearable health devices

Link. <https://www.electronics-foru.com/electronics-projects/>

heart-rate-monitor-wrist-band-reference-design

OEM Brand. Analog Devices (ADI)

Wearable sensor unit

Wearable sensors need to be compact, energy-efficient, and operate continuously across health, fitness, safety, and environmental applications. This reference design supports those needs through a watch strap form factor that enables



interaction with motion, environmental, and battery data. A microcontroller with embedded sensor fusion reduces processing load, supporting motion tracking, environmental sensing, and context-aware applications.

Data can be streamed and configured through mobile apps and BLE interfaces, while on-board cryptography secures information. By combining sensors, indicators, power management, cross-platform app support, and regulatory compliance, this design provides a ready-to-deploy wearable platform.

Applications. Suitable for wearable motion sensing

Link. <https://www.electronics-foru.com/electronics-projects/wearable-sensor-unit-reference-design>

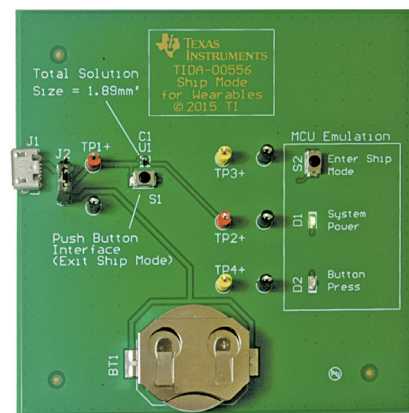
OEM Brand. STMicroelectronics

Ship mode for wearables

Battery drain during shipping, storage, or inactivity is a recurring challenge in portable electronics. This reference design addresses the issue by using a load switch to put the device into 'ship mode,' keeping it in an ultra-low-power state.

It enables controlled turn-on to prevent current surges, minimises leakage when off, and maintains very low power use when active. With a low-resistance MOSFET reducing voltage drop and a fixed slew rate preventing supply droop, the design remains compact, component-efficient, and power-conscious. The result is longer battery life, smoother out-of-box operation, and scalability across many wearable and portable devices.

Applications. Useful for smartwatches, navigation tools, industrial wearables, and other low-power electronics



Link. <https://www.electronics-foru.com/electronics-projects/industry-grade-projects/load-switch-based-ship-mode-reference-design-for-wearables>

OEM Brand. Texas Instruments

To conclude

From compact health monitors to energy-conscious power solutions, these reference designs show how much complexity can be reduced with ready-to-use platforms. Instead of starting from scratch, engineers can build devices that are compact, reliable, and efficient, while meeting the growing demands of health, fitness, safety, and industrial applications. **EFY**

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